

Video Transcript

Video 5: Housing Databases in Containers (06:31)

Given that we want to work with multiple databases, the process that we went through to install and configure set up MySQL is quite a lengthy one, and you can imagine that doing that in an organizational context would be even more so. So, what we would like to do is to take this idea of containers and containerize our databases so that they can run on Docker. That is, we want to be able to run this database on top of a platform and be able to access it through our machine as if that were something we went through and configured manually. That is very much the idea of containers.

Now, you could do this yourself. You could configure the application or the database in the application to be able to do that, so that later on if you wanted to reconstruct what you did, it would be easy to do so. And your teammates would certainly appreciate you doing that as well. Now, luckily for us, this idea of being able to create a registry of containers of images that we could leverage is something that has already been done. And if we navigate to this registry Docker Hub, we would be able here to search for anything we'd like. And as you can see there, I have typed in already MySQL and you can see there that that is the very first head.

You can see also that this is an official image, that it has over 10 million downloads, and that if I selected this, I would be able to see more of the documentation, the way in which I could run this, the different versions that are available and so on. Now, we could do the same thing for the other databases that we'd want to explore Mongo, you can see there's several versions. You can see Redis and you can see Cassandra as well. And so, you can see there that instead of building this container images ourselves, we can simply pull them from this registry.

The idea being that we run that, we load it onto our container, and then we access it from our local environment. Now, at this point, if you have not installed Docker, if the machine that you're in doesn't have it, go ahead and install it. Once you do so, go ahead and open up from your menu the dashboard, and once you do, you should see something like the following. As you can see here for me, I have no containers running. This is a clean environment. Now, in order to be able to load that image from the registry onto a container and then run on top of a platform, we need to write a command.

The general pattern of that command is the following. Note there the two braces that I have on top of the command as this being the minimal that you can do. That is `docker run`, and then the name of your image. And in our case, we're going to add more arguments. As you can see here, we're going to specify a port, and we're going to be recreating what we did before for MySQL, except this time we're going to be doing it using a container. We're going to go through all of the configuration that we did by hand, but we're going to do it here once again on Docker.

So, you can see there that we are writing `Docker run`, then the port, the container name, an environment variable that we want to pass in that sets the password. We're going to run this in a detached way. And then the image that we're going to be creating or we're going to be using is that of MySQL. So, as you can see here, I have my container dashboard and I'm going to go ahead and open up the terminal. And then I'm going to go ahead and paste that command that I had on my slides. And as you can see here, we have it once again, the same one `Docker run`, the port, the name, the variables that we're passing in and the image that we're using.

So, let's go ahead and run this. And as you can see there, that was very fast. And the reason for that is because I already had cached some of the images on my machine and this happens by default. You don't have to do anything to be able to get that. In fact, after the first time that you run it, your machine caches them for you. So, This is why it was faster. I'm going to go ahead now, and delete the images and you will see how much slower that is. And at the same time, I'm simply going to delete my container here, and this is one of the advantages.

If you think about everything we did before, in this case, we can simply get rid of everything we did by simply touching delete here or clicking on 'Delete'. And you can see there that is now gone. Now, I have just deleted the images that I had cached, and now I'm going to go ahead and run that command again. And as you can see there, it couldn't find it. It now is pulling out all of the different pieces that are necessary for that container. And as you can see there, that would take longer. Now, depending on the speed of your connection, this would be very fast as mine, I happen to be in a fast connection or it might take some more time.

But ultimately you can see here that the result is that we get a container. Note that there are a few options here. You can open it up in the browser and, that makes sense or not, depending on the type of container that you're running, you could step into the container, into the command line interface, you could stop the container, restart it, or simply delete it. If we open this up, you can see here some of what was executed, you can take a look at the logs, you could inspect, you

can see here more details about the environment, the mounts, the ports, and as well take a look at the stats. So, as you can see, this approach gives us tremendous speed. We can readily, in one line be able to recreate our entire development environment for the database, we can pull that from the registry, put it into a container and access it through our machine.

